



AgroLink Hub

## AgroLink Hub

Connect. Share. Sell. Grow Together.

FINAL PROJECT REPORT

# AgroLink Hub

Community Marketplace for Producers, Buyers and Creators

### Project vision

A practical platform for farmers, small businesses, buyers and creators to connect through community posts, seller pages, products, chat, orders, reviews, notifications and support.

### Farmers & producers

Share updates and sell directly

### Small businesses

Publish products and manage orders

### Buyers & creators

Discover, chat, review and engage

REPORT OVERVIEW

# Contents and Update Summary

This version rebuilds the report into a clean, print-ready PDF. The content is rewritten in a more natural style, the cover is redesigned with a high-resolution AgroLink theme background, and each page uses controlled spacing to avoid overlapping text or line overflow.

|                                               |                                                       |                                              |                                                 |
|-----------------------------------------------|-------------------------------------------------------|----------------------------------------------|-------------------------------------------------|
| <div>16</div> <div>total designed pages</div> | <div>30</div> <div>backend controllers reviewed</div> | <div>243</div> <div>backend Java files</div> | <div>126</div> <div>frontend source files</div> |
|-----------------------------------------------|-------------------------------------------------------|----------------------------------------------|-------------------------------------------------|

| Section                    | What it explains                                                                                                              |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Product summary and vision | The problem AgroLink solves and the value it gives to farmers, small sellers, buyers, creators, and admins.                   |
| Architecture and stack     | React frontend, Spring Boot backend, PostgreSQL persistence, WebSocket/STOMP realtime delivery, security, and media handling. |
| Feature scope              | Social feed, marketplace, seller workspace, cart, orders, chat, notifications, calendar, support, admin, and assistant.       |
| Operations and final notes | Build checks, deployment readiness, safety notes, performance warnings, final completed scope, and handover notes.            |

Main editing focus

Rewritten unnatural wording in a cleaner human style, removed references to artificial/generated visuals, corrected the security section to application-level access control, and added a clearer product vision section for the site.

EXECUTIVE SUMMARY

# What AgroLink Hub Is

AgroLink Hub is a full-stack social commerce platform for local communities, farmers, producers, small businesses, buyers, creators, and admins. It brings the daily engagement of a social network together with marketplace selling, order handling, and support workflows.

|                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>One place for discovery and selling</b></p> <p>Sellers do not need separate tools for audience building, chat, product listing, orders, reviews, and support. The platform keeps those workflows in one account workspace.</p> | <p><b>Built around trust</b></p> <p>Profiles, posts, comments, reviews, chat, seller pages, and visible activity help buyers understand who they are buying from before placing an order.</p>                                         |
| <p><b>Useful beyond farming</b></p> <p>The product is suitable for farmers and local producers, but the same flow also supports small shops, creators, handmade sellers, food suppliers, and community services.</p>                 | <p><b>Academic and practical value</b></p> <p>The project demonstrates a realistic full-stack implementation: authentication, roles, REST APIs, realtime messaging, persistent cart, orders, notifications, and admin moderation.</p> |

## Core statement

AgroLink Hub connects community engagement with direct commerce: users can discover people, follow updates, chat, buy products, track orders, review sellers, and ask for support without leaving the platform.

PRODUCT VISION

# The Site Direction and Expected Impact

The long-term direction is to become a trusted local digital marketplace where community relationships and buying decisions support each other. The platform should feel familiar like a social network, but still provide the structured tools needed for real selling.

|                                                                                                                                                                                             |                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Fairer producer reach</b></p> <p>Farmers and small producers can promote harvests, publish products, answer buyer questions, and reduce total dependence on intermediaries.</p>       | <p><b>Community-first commerce</b></p> <p>Buyers do not only see product cards. They see seller stories, updates, reviews, comments, and real conversations that make a purchase feel safer.</p>                       |
| <p><b>Practical seller operations</b></p> <p>Business and farmer accounts can manage seller pages, products, received orders, order status, reviews, and analytics from one workspace.</p>  | <p><b>Learning and local knowledge</b></p> <p>Education and news-oriented posts can make the feed useful for farming tips, price awareness, product quality, safety updates, and community learning.</p>               |
| <p><b>Admin and support confidence</b></p> <p>Admin tools, reporting, moderation, support tickets, and role workflows help the platform stay controlled as users and transactions grow.</p> | <p><b>Completed product direction</b></p> <p>The current project presents a unified community commerce experience with social discovery, seller pages, direct buying, chat, reviews, support, and admin oversight.</p> |

Vision sentence

The best version of AgroLink is not only a selling app. It is a community trust layer for local commerce.

USERS AND ACCESS

# Target Users and Role-Based Workspaces

The product uses different roles so each user sees the correct tools. Normal users focus on social and buying workflows, sellers manage products and orders, and admins handle support, reports, and moderation.

| Role          | Primary workspace and access                                                                               |
|---------------|------------------------------------------------------------------------------------------------------------|
| ROLE_USER     | Feed, marketplace browsing, cart, buyer orders, chat, support, calendar, profile, and settings.            |
| ROLE_CREATOR  | Social content creation, community reach, marketplace buying, chat, support, and profile management.       |
| ROLE_BUSINESS | Business page, product publishing, seller order management, reviews, and analytics.                        |
| ROLE_FARMER   | Farmer seller tools, produce/product listing, seller orders, reviews, and analytics.                       |
| ROLE_ADMIN    | Admin dashboard, user groups, moderation reports, support tickets, safety actions, and platform oversight. |

## User value flow

|                                                                                                                                                                      |                                                                                                                                                                         |                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Buyer</b><br><br>Discovers sellers through social posts or marketplace search, asks questions through chat, adds products to cart, checks out, and tracks orders. | <b>Seller</b><br><br>Builds credibility through posts and profile activity, publishes products, receives orders, updates status, and responds to reviews/support needs. | <b>Admin</b><br><br>Monitors user groups, reports, support requests, business accounts, farmers, creators, and admin-level platform activity. |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|

TECHNOLOGY STACK

# Frameworks, Libraries, and Runtime Tools

The project stack was checked from the uploaded source package. The frontend is a React single-page app, while the backend is a Spring Boot application with REST APIs, security, persistence, mail, and realtime messaging.

| Layer    | Technology                           | Purpose                                                                                                 |
|----------|--------------------------------------|---------------------------------------------------------------------------------------------------------|
| Frontend | React 18.3.1                         | Component-based pages, shared dashboard UI, auth screens, social, commerce, and platform workflows.     |
| Frontend | Vite 6.3.5                           | Local development server, asset bundling, and production frontend build.                                |
| Routing  | React Router DOM 6.30                | Public routes, protected app shell routes, and role-based route guards.                                 |
| HTTP     | Axios 1.8                            | Central API communication with credentials and refresh-aware service calls.                             |
| Realtime | @stomp/stompjs 7.1                   | WebSocket/STOMP messaging for chat, typing, presence, and notification behavior.                        |
| Backend  | Java 25, Spring Boot 4.0.2           | REST APIs, Spring Security, JPA, WebSocket, mail, validation, and MVC resource handling.                |
| Database | PostgreSQL                           | Relational persistence for accounts, content, marketplace, cart, orders, chat, support, and admin data. |
| Security | Spring Security, OAuth2 Client, JJWT | JWT auth, Google OAuth2, OTP flows, protected APIs, and role-specific access.                           |

Runtime configuration note

Environment values such as database credentials, mail credentials, JWT secret, OAuth2 settings, cookie options, CORS origins, server port, and upload path should remain outside public code and be configured per environment.

## ARCHITECTURE

# High-Level System Design

The codebase is split into a React frontend and Spring Boot backend. Both sides follow a domain layout around platform, social, and business modules, which makes a feature easier to trace from UI page to API endpoint to service and entity.

|                                                                                                                                                                                                      |                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>React client</b><br><br>Renders landing, auth, onboarding, protected app shell, feed, profile, friends, chat, marketplace, cart, orders, business, analytics, calendar, support, and admin pages. | <b>Spring Boot backend</b><br><br>Controllers expose REST endpoints, services contain workflow rules, repositories persist through JPA, and security filters protect authenticated APIs. |
| <b>Realtime channel</b><br><br>WebSocket/STOMP is configured for chat messages, typing indicators, presence updates, and realtime notification behavior.                                             | <b>Media delivery</b><br><br>Uploaded media is stored through the backend and served through a configured resource path, with upload limits set for larger images and videos.            |

## Typical request flow

- The browser calls the backend through the shared Axios client.
- Authentication details are sent through headers or cookies, depending on the flow.
- Spring Security authenticates the request and checks the route permissions.
- Controllers delegate work to services, services call repositories, and PostgreSQL stores the data.
- React pages receive DTO-style responses and update the dashboard state, UI cards, tables, or realtime views.



## REPOSITORY STRUCTURE

# Folder and Module Organization

The uploaded repository keeps frontend, backend, uploads, and documentation separated. The strongest maintainability point is the mirrored domain structure across frontend and backend modules.

| Path            | Meaning                                                                                                                                          |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Lisharefrontend | React/Vite application with assets, modules, shared UI, CSS, services, route definitions, and page-level components.                             |
| Lisharebackend  | Spring Boot API application with controllers, services, repositories, entities, security, configuration, schema SQL, and mail/WebSocket support. |
| uploads         | Local development storage for uploaded media. Production should use persistent storage or cloud object storage.                                  |
| docs            | Documentation output folder for the project report and supporting materials.                                                                     |

|                                                                                                                                         |                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Frontend modules/platform</b><br>Authentication, app routes/store, layouts, common UI, user profile/settings, support, and calendar. | <b>Frontend modules/social</b><br>Feed, posts, friends, follow, chat, notifications, chatbot services, and social pages.                                                |
| <b>Frontend modules/business</b><br>Marketplace, products, cart, orders, business pages, reviews, support, analytics, and admin views.  | <b>Backend modules</b><br>platform, social, and business packages contain matching controllers, services, entities, repositories, security, storage, and configuration. |



## FRONTEND

# Pages, Routes, and UI Behavior

The frontend is a single-page application. Public routes handle landing and account flows, while authenticated pages render inside the app shell with navigation, page metadata, notifications, user context, and consistent layout patterns.

| Area     | Pages                                                                           | Purpose                                                                           |
|----------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Auth     | Landing, login, signup, verify OTP, forgot password, OAuth callback, onboarding | Account creation, verification, session start, recovery, and first-time setup.    |
| Social   | Home feed, profile, friends, chat, notifications                                | Community feed, user identity, relationships, realtime messaging, and alerts.     |
| Commerce | Marketplace, cart, orders, business page, analytics                             | Product browsing, shopping, seller publishing, order tracking, and insights.      |
| Platform | Calendar, support, settings                                                     | Planning, help desk, account media, email/password changes, and user preferences. |
| Admin    | Admin dashboard and admin subroutes                                             | User groups, role workflows, reports, support tickets, and moderation.            |

## Design system notes

- Shared dashboard UI components support cards, buttons, modals, badges, avatars, stat cards, tabs, and layout behavior.
- The visual palette follows AgroLink green, yellow, teal, dark green surfaces, and restrained red only for destructive actions.
- Service files use centralized endpoint constants and the shared Axios client so API paths remain consistent across features.

## BACKEND

# API Responsibilities and Service Layer

The backend contains controllers, services, repositories, entities, security filters, WebSocket configuration, mail support, validation, schema management, and upload handling.

| Module                 | Main responsibilities                                                                                                          |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| platform/auth          | Register, login, refresh, logout, OTP verification, forgot password, and Google OAuth2.                                        |
| platform/user          | Profiles, account updates, media, blocking, settings, and public user data.                                                    |
| platform/security      | Security filter chain, JWT authentication, route permissions, and role enforcement.                                            |
| social/post            | Posts, feed, saved posts, reports, media, polls, audience metadata, and moderation support.                                    |
| social/comment/story   | Comments, replies, reactions, story media, story views, expiry behavior, and story interactions.                               |
| social/chat            | Direct and group conversations, messages, attachments, reactions, seen state, presence, typing events, and WebSocket delivery. |
| business/product/page  | Marketplace products, seller pages, availability, stock, public browsing, and business/farmer profiles.                        |
| business/cart/order    | Cart persistence, quantity update, checkout, buyer orders, seller received orders, cancellation, and status updates.           |
| business/admin/support | Admin user groups, moderation, reports, support tickets, review handling, replies, and platform stats.                         |

## Backend quality note

The service layer is important because it keeps business decisions out of React components and controller methods. That makes registration, checkout, order status updates, support replies, chat conversations, notifications, and moderation easier to test and maintain.

## DATABASE

# Persistence and Storage Model

PostgreSQL is used as the relational database. Spring Data JPA maps entities to tables, while schema files and patching logic keep local databases compatible as the feature set grows.

|                                                                                                                          |                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Users and auth</b><br>Users, roles, OTP fields, forgot-password records, profile media, account metadata, and blocks. | <b>Social graph</b><br>Follow, friends, notifications, saved posts, post reports, stories, story views, and story reactions. |
| <b>Content</b><br>Posts, poll votes, comments, comment reactions, reactions, shares, story media, and media URL fields.  | <b>Commerce</b><br>Business pages, products, cart items, orders, reviews, support questions, and order statuses.             |
| <b>Chat</b><br>Conversations, members, messages, seen states, attachments, presence, typing events, and reactions.       | <b>Assistant</b><br>Knowledge concepts with topic, keywords, and answer description for guided assistant responses.          |

## Storage behavior

- Uploaded files are exposed through the backend resource mapping path.
- The development project stores media locally in the uploads folder.
- Production should use persistent upload storage and a carefully configured maximum upload size.

## SECURITY

# Authentication, Authorization, and Access Control

The project uses application-level access control rather than database-native row-level security. Backend route permissions, service checks, frontend protected routes, and role-based navigation work together to keep workflows behind the correct permissions.

|                                                                                                                                                                                       |                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Authentication methods</b><br><br>Email/password login, signup OTP verification, forgot-password OTP flow, refresh token handling, logout, and Google OAuth2 login.                | <b>Token strategy</b><br><br>JWT access/refresh behavior is supported through headers or cookies depending on the auth flow and frontend session handling.                               |
| <b>Backend protection</b><br><br>Public routes are limited to auth, recovery, OAuth2, selected public resources, uploads, and landing/assistant flows. Admin APIs require admin role. | <b>Frontend protection</b><br><br>ProtectedRoute blocks logged-out users from app shell pages. RoleRoute blocks non-seller users from seller tools and non-admin users from admin tools. |

| Area           | Professional recommendation                                                                               |
|----------------|-----------------------------------------------------------------------------------------------------------|
| Secrets        | Keep database, mail, OAuth2, and JWT values outside git. Rotate real secrets if they were ever committed. |
| Cookies        | Use secure, same-site aware cookie settings in HTTPS production.                                          |
| CORS           | Restrict production CORS origins to trusted frontend domains only.                                        |
| Admin accounts | Create admin accounts through a controlled process, not through unrestricted public signup.               |

SOCIAL FEATURES

# Community Feed and Engagement

The social layer keeps the platform active between purchases. It helps sellers build trust and lets users share updates, learn from each other, ask questions, and stay connected.

|                                                                                                                                                                                       |                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Feed and posts</b><br><br>Feed loading, post creation, media URLs, audience metadata, feelings, location data, poll support, saved posts, shares, reports, and moderation support. | <b>Comments and reactions</b><br><br>Users can comment, reply, update and delete comments, react to posts, and react to comments.                    |
| <b>Stories</b><br><br>Temporary media stories, story feed viewing, reactions, replies, sharing, view tracking, and deletion.                                                          | <b>Friends and following</b><br><br>Search, follow/unfollow, follower/following counts, friend requests, accepts, rejects, cancels, and unfriending. |
| <b>Profiles</b><br><br>Public profile views, identity, profile media, cover images, posts, saved items, bio details, and account actions.                                             | <b>Notifications</b><br><br>Unread count, notification list, mark-read, read-all, delete, and clear-all behavior with compact dropdown design.       |

Content quality idea

Education and news category guidance can turn the feed into a useful community knowledge space instead of only a place to scroll. That aligns well with farming tips, safety updates, market awareness, and local business learning.

## BUSINESS FEATURES

# Marketplace, Cart, Seller Workspace, and Orders

The business side lets farmers and small businesses publish products while buyers purchase directly from them. Product discovery, cart persistence, checkout, order tracking, seller updates, reviews, and analytics connect into one workflow.

| Feature        | What it does                                                                                                               |
|----------------|----------------------------------------------------------------------------------------------------------------------------|
| Business pages | Business and farmer accounts can manage seller pages and public seller profiles.                                           |
| Products       | Sellers publish products with category, price, stock, availability, images, delivery methods, and seller details.          |
| Marketplace    | Buyers browse products, search/filter, inspect details, add to cart, and navigate to seller pages.                         |
| Cart           | Buyers update quantities, view totals, remove items, clear cart, and proceed to checkout with persisted backend cart rows. |
| Orders         | Checkout creates backend orders. Buyers track order history and sellers update received-order status.                      |
| Reviews        | Users can leave website or business-related reviews that support trust and discovery.                                      |
| Analytics      | Business and platform insights help sellers and admins understand product and order activity.                              |

## Order flow

- User discovers products from the marketplace or seller page.
- Product is added to a backend-persisted cart.
- Buyer reviews quantity, delivery method, and order total.
- Checkout creates order records and clears the cart.
- Orders page shows buyer order history or seller received orders depending on role.

## PLATFORM TOOLS

# Messaging, Support, Notifications, Calendar, and Assistant

These tools make the product more complete and easier to use. Users can communicate, plan, ask for help, receive updates, and get guided answers without depending only on manual navigation.

|                                                                                                                                                                                                              |                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Realtime chat</b><br><br>Direct and group conversation support, message APIs, image/video attachments, reactions, sent/delivered/read ticks, seen state, presence, typing events, and WebSocket delivery. | <b>Support center</b><br><br>Users create support questions and view admin responses. Admin support routes manage tickets and replies.                                                     |
| <b>Calendar</b><br><br>Events, reminders, farm tasks, business planning, birthdays, meetings, and visibility options.                                                                                        | <b>Admin command center</b><br><br>User groups, business users, farmers, creators, admins, moderation reports, support tickets, and platform oversight.                                    |
| <b>Notifications</b><br><br>Realtime and API-supported notification behavior gives users visible updates for actions and workflow changes.                                                                   | <b>Chatbot assistant</b><br><br>A knowledge-base assistant answers common questions about signup, login, marketplace, selling, posts, stories, orders, cart, support, roles, and security. |

## Why chat matters

For buyer-seller commerce, messaging is not only a social feature. It supports stock questions, price negotiation, delivery details, order confidence, and after-sale support.

## Marketplace trust and safety

AgroLink protects marketplace quality through reports, reviews, support requests, and admin moderation. If a fake seller, misleading product, scam-like activity, low-quality seller behavior, or repeated misuse appears, users can raise feedback and admins can review the issue, reply through support, and take account-level action. Business, farmer, and creator accounts can be warned or suspended when misuse is identified, which helps protect buyers, genuine sellers, and platform trust.



OPERATIONS

# Build, Testing, Deployment Readiness, and Final Notes

This closing page keeps the report focused on the completed project scope. It records the build, verification, configuration, and handover notes for the final submitted version.

| Area                | Readiness note                                                                                                                                 |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend build      | Run npm install and npm run build inside Lisharefrontend. Vite chunk-size warnings should be treated as performance notes, not build failures. |
| Backend build       | Run Maven wrapper or local Maven inside Lisharebackend with PostgreSQL and required environment values available.                              |
| Database            | Create the PostgreSQL database, configure DB_URL, DB_USERNAME, and DB_PASSWORD, and allow JPA/schema logic to prepare required tables.         |
| Production security | Use a strong JWT secret, HTTPS cookie settings, restricted CORS, configured mail credentials, and trusted OAuth2 redirect URLs.                |
| Media storage       | Local uploads are acceptable for development and demonstration. A real hosted deployment should use persistent storage for uploaded media.     |

## Final completed scope

|                                                                                                                                                                           |                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Account and roles</b><br><br>Signup, login, OTP flows, Google OAuth2, JWT session behavior, protected routes, seller roles, and admin role separation.                 | <b>Social workspace</b><br><br>Posts, comments, reactions, shares, stories, friends/following, profiles, notifications, and content discovery.                                             |
| <b>Commerce workflow</b><br><br>Business pages, farmer/seller pages, products, marketplace browsing, cart, checkout, buyer orders, seller orders, reviews, and analytics. | <b>Communication and support</b><br><br>Realtime chat, media messages, delivery/read ticks, presence, typing indicators, support tickets, admin replies, calendar, and assistant guidance. |

### Closure

Final summary: AgroLink Hub is a complete full-stack social commerce demonstration that connects community interaction with direct selling. The implemented scope combines authentication, role-based workspaces, social media, marketplace, cart, orders, chat, notifications, support, calendar, analytics, admin moderation, and a knowledge-base assistant.